



Use of I²C for Extension of GPIOs

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Raspberry Pi has a total of 26 pins on the GPIO header, out of which only 17 can be interfaced with external hardware. But to further extend the interfacing you can connect an additional microcontroller and increase the GPIO capability by any number. Even the regular LCD



Fig. 1: Author's prototype

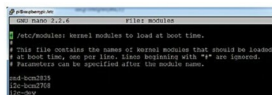


Fig. 2: Adding the lines in /etc/modules



Fig. 3: Editing the raspi-blacklist.conf file

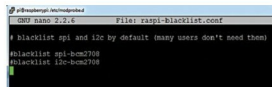


Fig. 4: Commenting the code

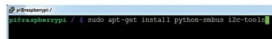


Fig. 5: Installing python-smbus and I2C tools

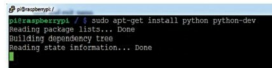


Fig. 6: Installing Python packages

panel, which eats away half a dozen of GPIOs, can straightaway be shifted to the additional microcontroller, consuming only two pins of Raspberry Pi to control all these countless GPIOs precisely yet very economically.

In July issue we had shown how to interface dot-matrix LED displays using I²C pins of Raspberry Pi and I/O expander MCP23017. Here we will further extend the GPIOs of the Raspberry Pi using I/O expander MCP23008 to interface multiple devices, such as digital temperature sensor DS18B20 and 16x4 LCD. The LCD will display the IP address, temperature, date and time.

Here I²C communication is used. I²C is a serial communications protocol which allows chips to swap data on the same bus. The I²C is just two wires, SDA and SCL (pin 3 and pin 5), which control all the I²C devices connected on Raspberry Pi computer.

System set up

To use I²C on Raspberry Pi, you need to enable a few things in Raspbian Wheezy operating system, as by default it is not enabled. This is a fairly easy process. First you need to edit the modules file using the command:

```
# sudo nano /etc/modules
```

Then add the following two lines (also refer Fig. 2):

```
i2c-bcm2708
i2c-dev
```

Save the file and exit nano. Next, open the raspi-blacklist.conf file by writing the following command (also refer Fig. 3):

```
# sudo nano /etc/modprobe.d/raspi-blacklist.conf
```

Comment out the code by putting # symbol at the beginning of the two lines as

shown below (also refer Fig. 4) and then save and exit.

```
#blacklist spi-bcm2708
#blacklist i2c-bcm2708
```

Now install the necessary software to manipulate the I²C bus using following commands (also refer Fig. 5):

```
# sudo apt-get update
# sudo apt-get install python-smbus
i2c-tools
```

Once the software is installed, reboot your Raspberry Pi so that these effects take place permanently. Also install Python and python-dev on Raspberry Pi using the following commands (also refer Fig. 6):

```
# sudo apt-get install python
python-dev
```

For testing whether the I²C is successfully enabled, refer article 'Interfacing Dot-Matrix LED With Raspberry Pi' published in July issue.

Circuit and working

Make the connections as shown in Fig. 7 to interface 16x4 LCD (LCD1) and temperature sensor DS18B20 (IC2). Fig. 8 shows pin configuration of the temperature sensor.

Wire up the LCD panel as shown in Fig. 7.

The temperature sensor is connected directly to the Raspberry Pi whereas the LCD panel is connected through MCP23008. Apart from Ground and 5V, only two more wires, SDA and SCL, have gone to the MCP23008 chip as shown in Fig. 7.

Software

The program uses Adafruit_MCP230xx.py and Adafruit_I2C.py GPL libraries which are taken from Adafruit.com. The source code has two other

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com